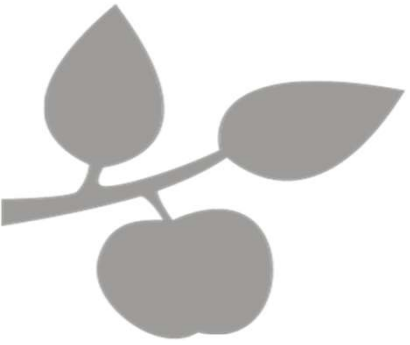




DM583

Data Mining

Unsupervised Outlier Detection

Authorized Adaption & Extension of the Lecture:

Outlier Detection

Dr. Arthur Zimek
University of Southern Denmark

Introduction

What is an Outlier?

Definition of Hawkins [Hawkins 1980]:

“An outlier is an observation which deviates so much from the other observations as to arouse suspicions that it was generated by a different mechanism”

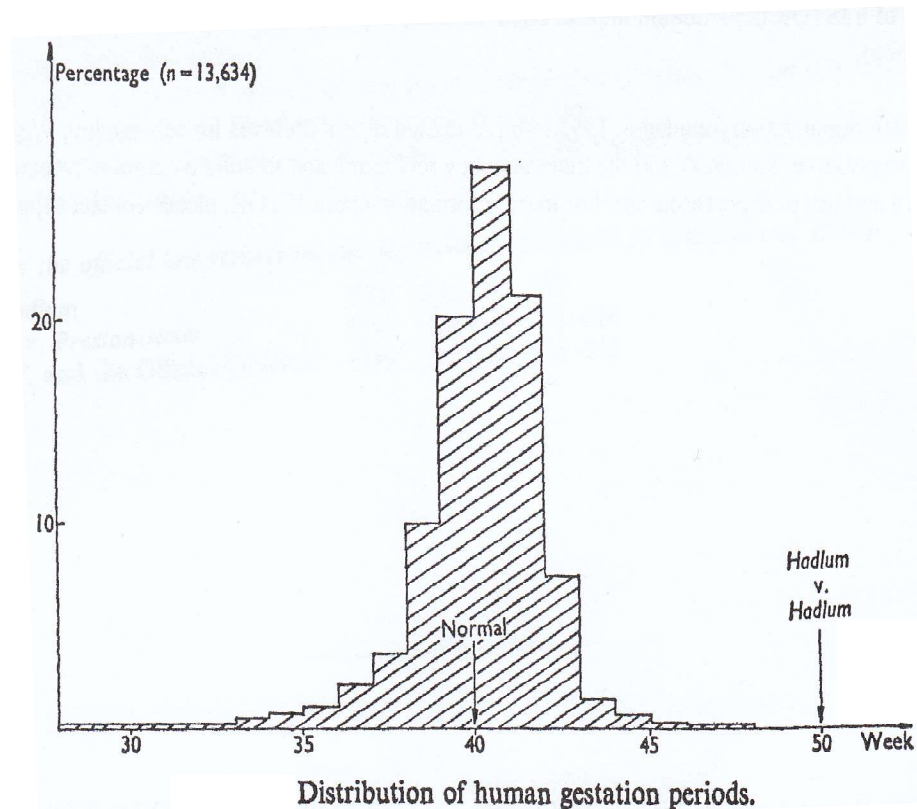
Statistics-Based Intuition

- Normal observations follow a “generating mechanism”, e.g. some given statistical process
- Abnormal observations deviate from this generating mechanism

Introduction

- Example: Hadlum vs. Hadlum (1949) [Barnett 1978]

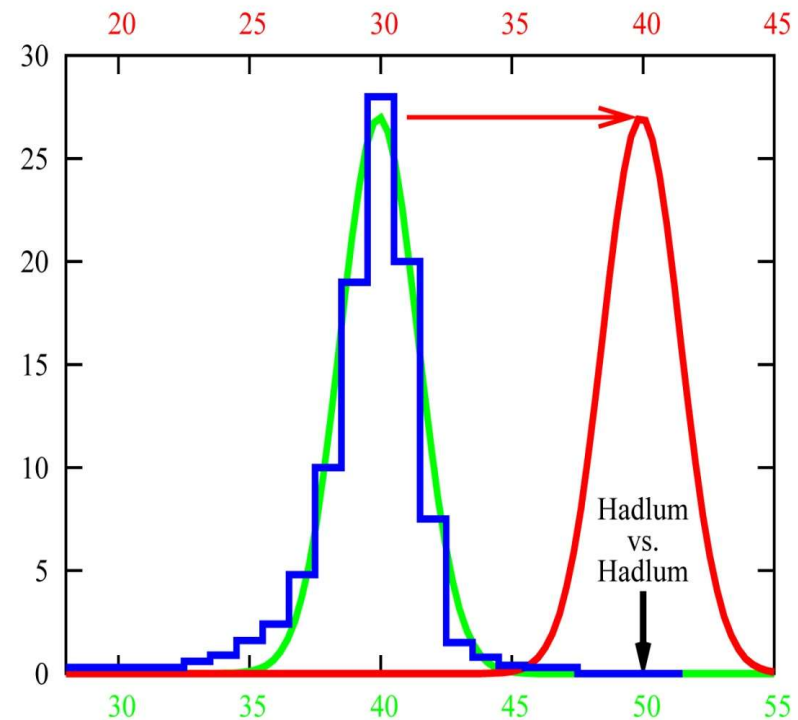
- The birth of a child to Mrs. Hadlum happened 349 days after Mr. Hadlum left for military service.
- Average human gestation period is 280 days (40 weeks).
- Statistically, 349 days is an outlier.



Introduction

- Example: Hadlum vs. Hadlum (1949) [Barnett 1978]

- blue: statistical basis (13634 observations of gestation periods)
- green: assumed underlying Gaussian process
 - Very low probability for the birth of Mrs. Hadlum's child for being generated by this process
- red: Mr. Hadlum's assumption (another Gaussian process is responsible for the observed birth; gestation period starts later)
 - Under this assumption, the gestation period has an average duration, and the specific birthday has highest-possible probability



Introduction

- Sample applications of outlier detection
 - Fraud detection
 - Purchasing behavior of a credit card owner usually changes when the card is stolen
 - Abnormal buying patterns can characterize credit card abuse
 - Medicine
 - Unusual symptoms or test results may indicate potential health problems of a patient
 - Whether a particular test result is abnormal may depend on other characteristics of the patients (e.g. gender, age, ...)

Introduction

- Sample applications of outlier detection (cont.)
 - Sports statistics
 - Various parameters are recorded for players in order to evaluate the players' performances
 - Outstanding (in a positive as well as a negative sense) players may be identified as having abnormal parameter values
 - Sometimes, players show abnormal values only on a subset of the recorded parameters
 - Detecting measurement errors
 - Data derived from sensors (e.g., in a given scientific experiment) may contain measurement errors
 - Abnormal values could provide an indication of a measurement error
 - Removing such errors can be important in other data analysis tasks
 - ...

Introduction

- General application scenarios
 - **Supervised Scenario**
 - In some applications, training data with normal and abnormal observations are provided
 - There may be multiple normal and/or abnormal classes
 - Often, the classification problem is highly imbalanced
 - **Semi-Supervised Scenario**
 - In some applications, only training data for one of the classes (usually the “normal” class) are available
 - one-class classification problem
 - **Unsupervised Scenario**
 - In many applications there are no training data available

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 - one-class classification problem
 - **Unsupervised Scenario**
 - In many applications there are no training data available
- here, we focus on the *unsupervised scenario*

Introduction

- Main unsupervised outlier detection approaches:
 - **Statistical Approach**
 - Outliers are unusual observations according to an estimated (usually parametric) probability distribution
 - **Clustering-Based Approach**
 - Outliers are observations that do not match well any cluster, or they are those that constitute a relatively very small cluster
 - **Non-Parametric Density-Based Approach**
 - Outliers are observations with unusual (absolute or relative) density

Introduction

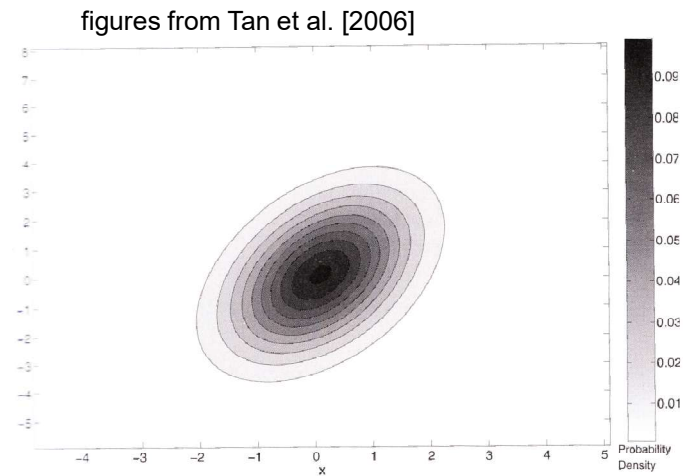
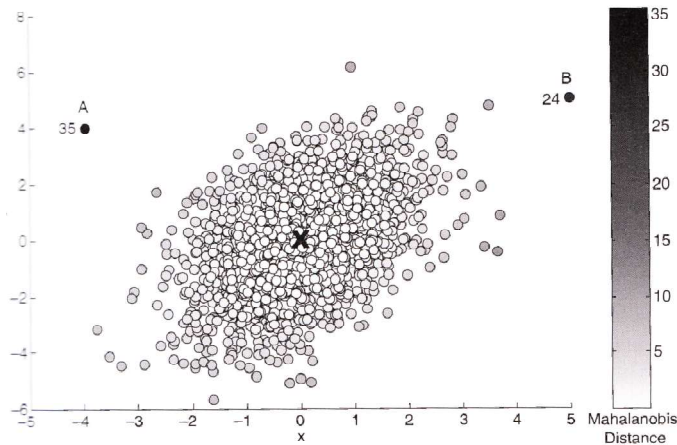
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Statistical Tests

- General idea
 - Given a certain type of statistical distribution (e.g., Gaussian)
 - Compute the parameters assuming all data points have been generated by such a statistical distribution (e.g., mean and standard deviation)
 - Outliers are points that have a low probability to be generated by the overall distribution (e.g., deviate more than 3 times the standard deviation from the mean of a Gaussian)
- Basic assumption
 - Normal data objects follow a (known) distribution and occur in a high probability region of this model
 - Outliers deviate strongly from this distribution

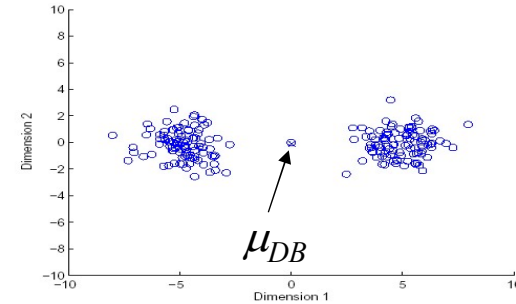
Statistical Tests

- Many different tests are available, differing in:
 - Type of data distribution (e.g. Gaussian)
 - Number of variables, i.e., dimensions of the data objects (univariate/multivariate)
 - Number of distributions (mixture models)
 - ...
- Example (2D):



Statistical Tests

- Main Problems:
 - Parametric Assumption
 - A specific distribution must be assumed, but the assumption may not hold true for the data at hand. **True distribution is rarely known in practice**
 - For example, simple unimodal distributions that are well-studied and easy to apply / manipulate do not represent many real-world scenarios
 - On the other hand, non-parametric methods like histograms are usually not accurate and/or computationally feasible in high-dimensional data



Statistical Tests

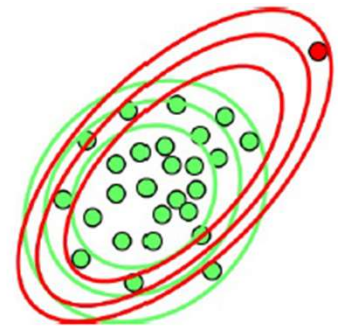
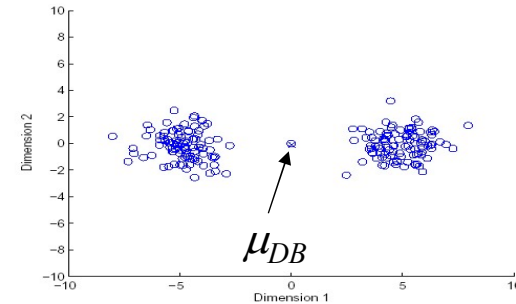
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- Robustness

- Distribution parameters (e.g. mean and standard deviation) may be very sensitive to outliers: **outliers may *mask* themselves as false negatives or *swamp* inliers as false positives by affecting distribution estimates**
 - Sophisticated (robust) distribution estimation techniques are required



Statistical Tests

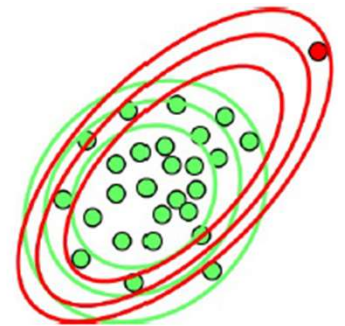
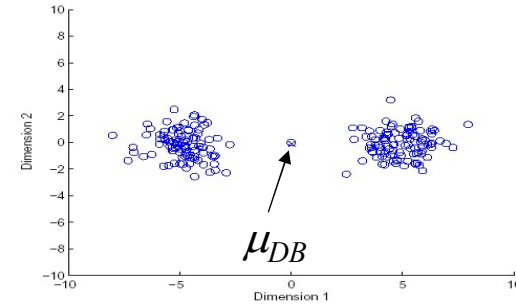
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- Robustness

- Distribution parameters (e.g. mean and standard deviation) may be very sensitive to outliers: **outliers may *mask* themselves as false negatives or *swamp* inliers as false positives by affecting distribution estimates**
 - Sophisticated (robust) distribution estimation techniques are required
 - **Estimation in higher dimensional data can be very difficult, if possible**



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 - Outliers are observations with unusual (absolute or relative) density

Clustering-Based Approaches

- Main Problems:
 - Effectiveness depends highly on the clustering method used (**they may not be optimized for outlier detection**)
 - Computational cost may be prohibitive depending on the clustering method used (need to 1st find clusters, then outliers)
 - More sophisticated clustering algorithms, e.g., able to model the notion of noise and automatically determine the number of clusters in data, are often computationally more expensive

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 - **Non-Parametric Density-Based Approach**
 - Outliers are observations with unusual (absolute or relative) density
- In the following, we provide a few classic examples of the non-parametric *density-based* approach

Non-Parametric Density-Based Global Approaches

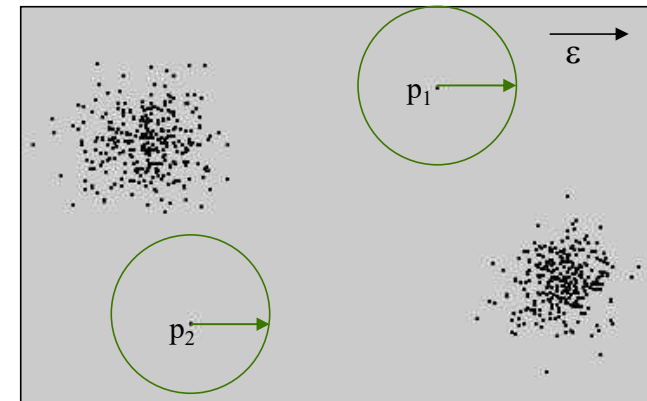
- General idea
 - Perform a non-parametric density estimate
 - Take the density of an observation as an absolute (global) measure of “inlierness”
 - Measure of outlyingness can be derived, e.g., as its inverse or complement
- Basic assumption
 - The density around a normal data observation (an inlier) is high
 - The density around an outlier is low

Precursor of Non-Parametric Density-Based Global Approaches

- DB(ε, π)-Outliers (Distance-Based Approach)

- Basic model [Knorr and Ng 1997]

- Given a radius ε and a percentage π
 - A point p is considered an outlier if at most π percent of all other points have a distance to p less than ε



$$\text{OutlierSet}(\varepsilon, \pi) = \{p \mid \frac{\text{Card}(\{q \in \text{DB} \mid \text{dist}(p, q) < \varepsilon\})}{\text{Card}(\text{DB})} \leq \pi\}$$

range-query with radius ε

$\text{Card}(\cdot)$: Cardinality
 DB : Database (dataset)
 $\text{Card}(\text{DB})$: No. of data points

- How can we modify the above definition, so we get rid of a parameter and at the same time provide *outlier scores* for each data object, rather than a binary classification (inliers vs outliers)??

Non-Parametric Density-Based Global Approaches

- Outlier scoring based on k NN distances (also deemed a **distance-based approach**)
 - One of the simplest approaches
 - **Density** roughly estimated as inverse of distance(s) of a point to its k -nearest neighbor(s)

Two general models:

- **KNN Method**: Take the k NN distance of a point as its outlier score
- **Weighted KNN Method**: Aggregate (average) the distances of a point to all its 1NN, 2NN, ..., k NN as an outlier score

Result: “outlier scores”

Non-Parametric Density-Based Global Approaches

- Example (simple KNN Outlier):

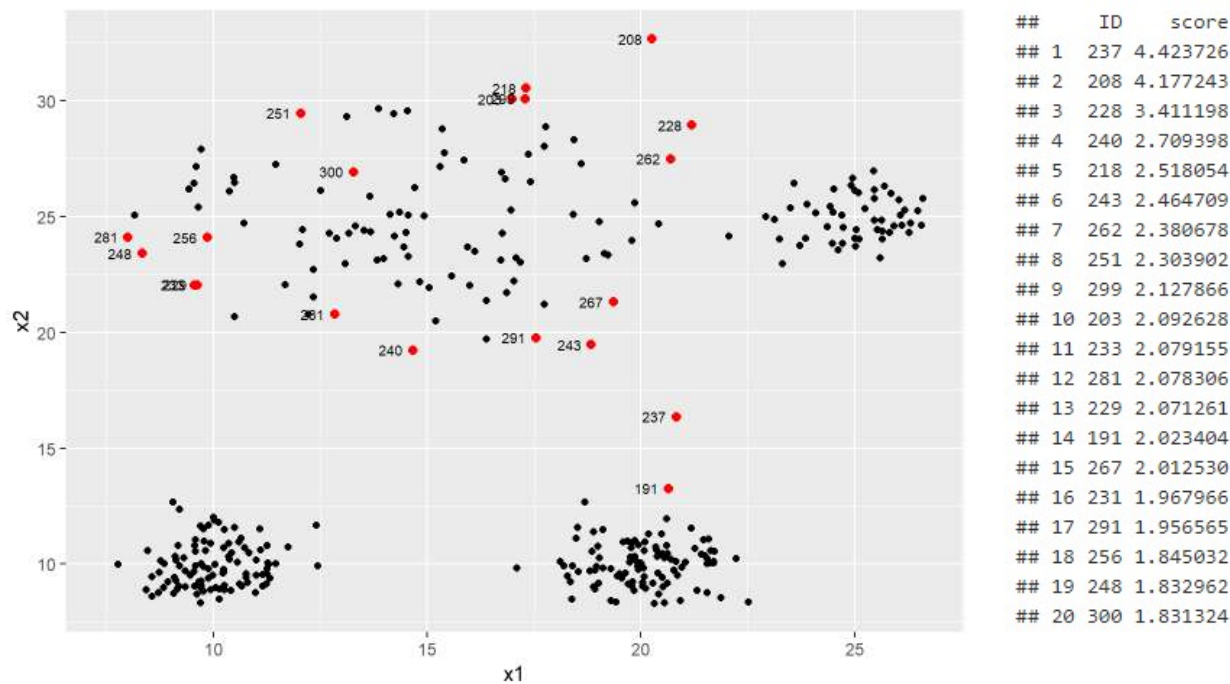


Fig. 2. KNN Outlier ($k = 4$): red points are the top 20 outliers, with their IDs

Non-Parametric Density-Based Global Approaches

- Choosing the k Parameter:

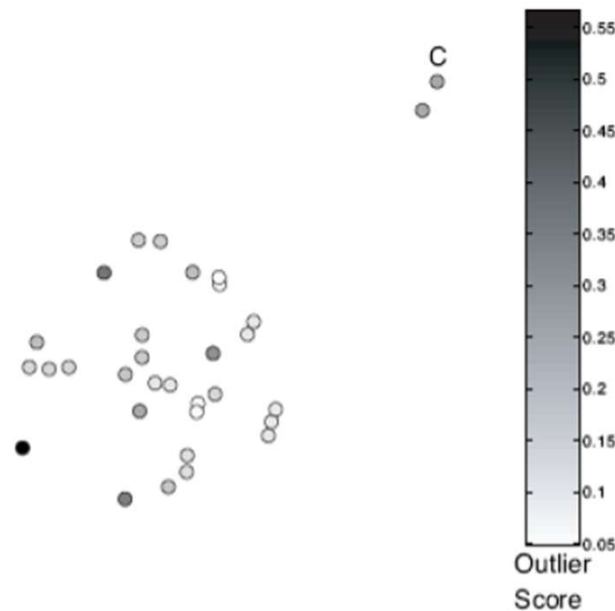


Figure 10.5. Outlier score based on the distance to the first nearest neighbor. Nearby outliers have low outlier scores.

(Figures from Tan et al. [2006].)

Non-Parametric Density-Based Global Approaches

- Choosing the k Parameter:

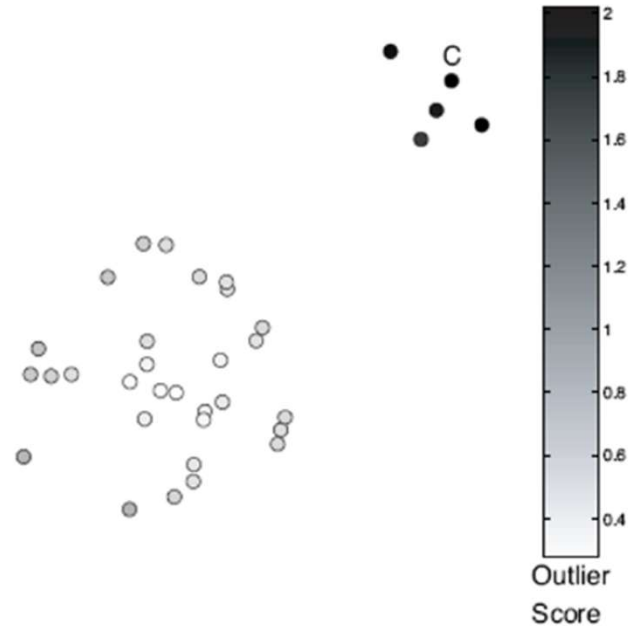


Figure 10.6. Outlier score based on distance to the fifth nearest neighbor. A small cluster becomes an outlier.

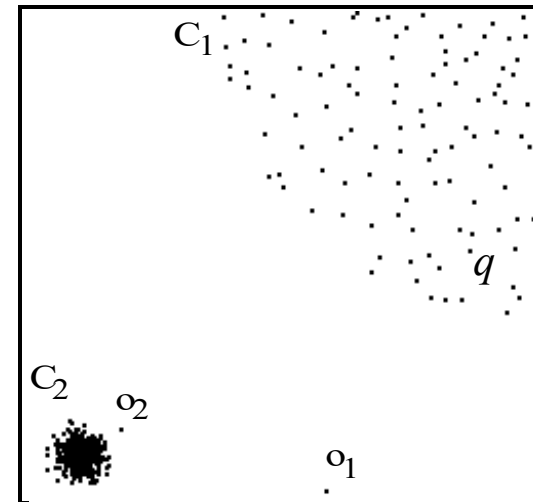
(Figures from Tan et al. [2006].)

Non-Parametric Density-Based Local Approaches

- General idea
 - Compare the density around a point with the density around its local neighbors
 - The relative density of a point as contrasted to its neighbors' is computed as an outlier score
 - Approaches also differ in how to estimate density
- Basic assumption
 - The density around a normal data observation is similar to the density around its neighbors
 - The density around an outlier is considerably different to the density around its neighbors

Non-Parametric Density-Based Local Approaches

- Local Outlier Factor (LOF)
- Motivation:
 - How to detect observations that are outliers relative to a certain local subset of the data
 - Global models have problems, particularly when there are regions with different densities
 - Example
 - Outliers based on kNN-distance
 - » kNN-distances of observations in C_1 (e.g. q) are larger than the kNN-distance of o_2
 - Solution: consider relative density



Non-Parametric Density-Based Local Approaches

- Draft LOF Model (Oversimplistic, for Pedagogic Purposes – See Next Slide)

- “Density” estimate for observation p
 - Inverse of the average distances from the k NNs of p

$$lrd_k(p) = 1 / \left(\frac{\sum_{o \in kNN(p)} dist(p, o)}{Card(kNN(p))} \right)$$

- (Draft) LOF score of observation p
 - *Average ratio of lrd's of neighbors of p and lrd of p*

$$LOF_k(p) = \frac{\sum_{o \in kNN(p)} \frac{lrd_k(o)}{lrd_k(p)}}{Card(kNN(p))}$$

Non-Parametric Density-Based Local Approaches

– Original LOF Model:

- Uses *reachability distance* rather than simple distance
 - This has the effect of reducing statistical fluctuations within clusters

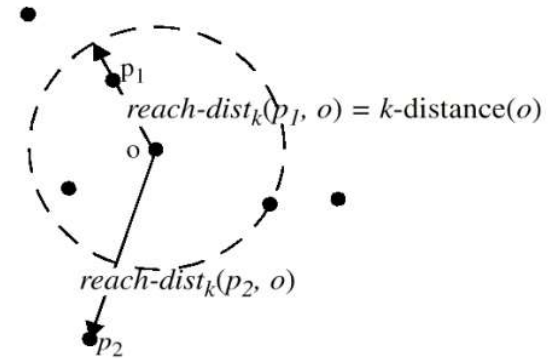
$$reach_dist_k(p, o) = \max\{k_distance(o), dist(p, o)\}$$

- Local reachability distance (lrd) of observation p
 - Inverse of the average reach_dists from the k NNs of p

$$lrd_k(p) = 1 / \left(\frac{\sum_{o \in kNN(p)} reach_dist_k(p, o)}{Card(kNN(p))} \right)$$

- Local Outlier Factor (LOF) of observation p
 - Average ratio of lrd's of neighbors of p and lrd of p

$$LOF_k(p) = \frac{\sum_{o \in kNN(p)} \frac{lrd_k(o)}{lrd_k(p)}}{Card(kNN(p))}$$

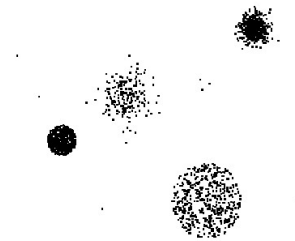


$k_distance(o)$ or $d_k(o)$: distance from point o to its k^{th} nearest neighbor ($k = 3$ in the example above)

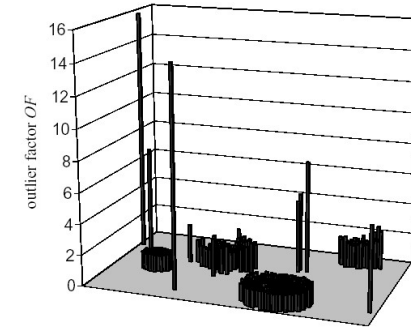
Non-Parametric Density-Based Local Approaches

– Properties:

- $\text{LOF} \approx 1$: point is in a cluster (region with homogeneous density around the point and its neighbors)
- $\text{LOF} \gg 1$: point is an outlier



Data set



LOFs ($k = 40$)

– Choice of k:

- Specifies the reference local data subset
- Works as a smoothing term of the density estimates

Non-Parametric Density-Based Local Approaches

- Example (LOF):

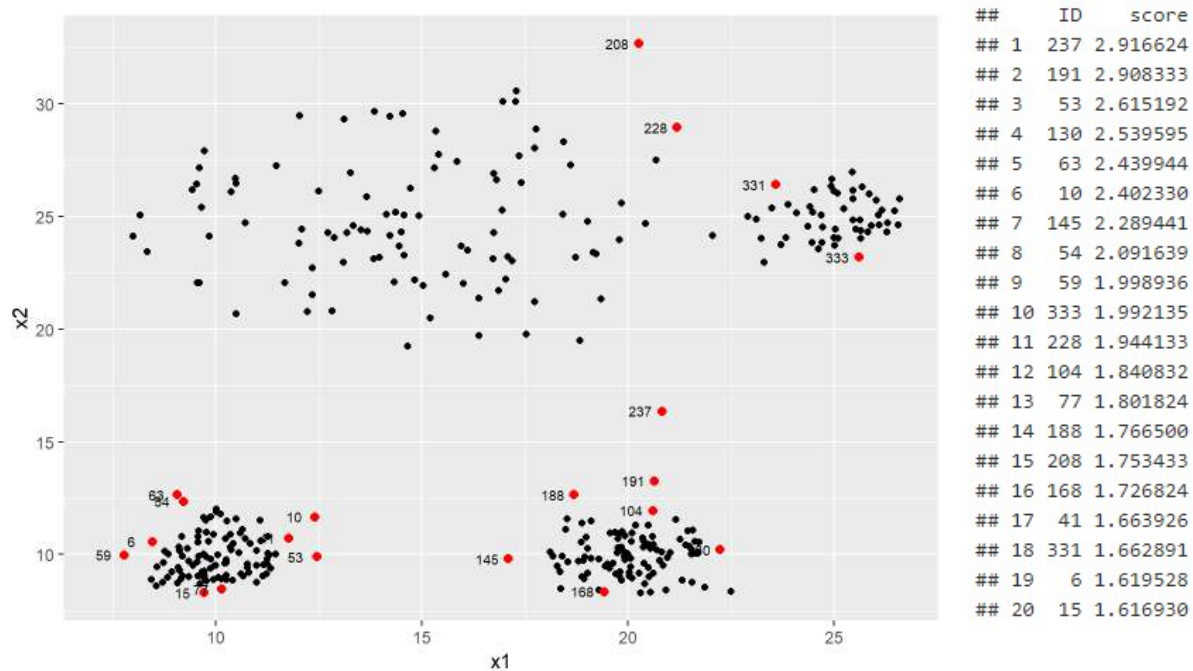
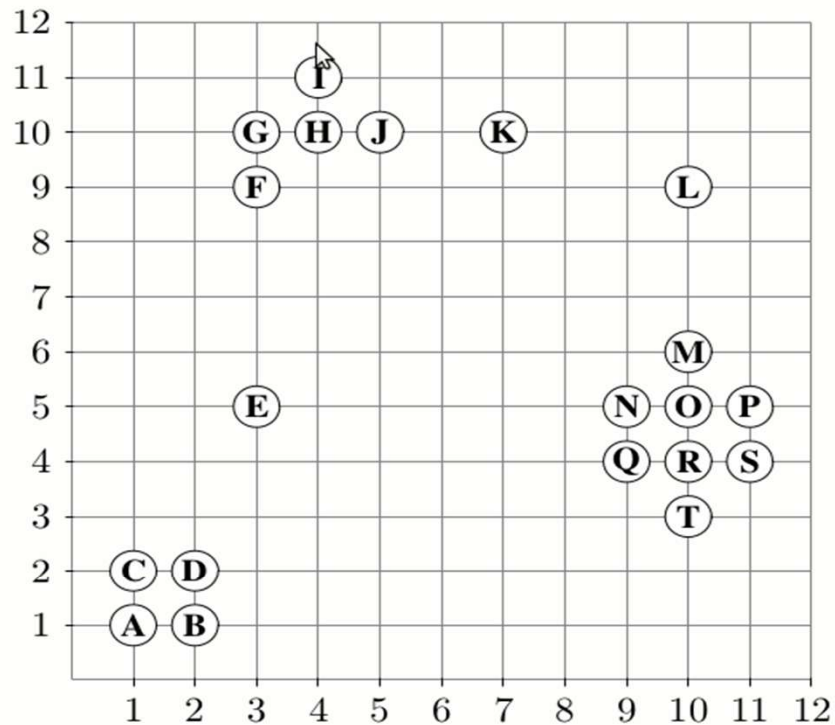


Fig. 4. LOF ($k = 4$): red points are the top 20 outliers, with their IDs

Exercise

Exercise 1-1 *Outlier Scores*



As distance function, use Manhattan distance $L_1(a, b) := |a_1 - b_1| + |a_2 - b_2|$.

Compute the following (without including the query point when determining the k NN):

Exercise (continued)

- LOF using $k = 2$ for the points E , K and O .
- LOF using $k = 4$ for the points E , K and O .
- k NN distance using $k = 2$ for all points.
- k NN distance using $k = 4$ for all points.
- aggregated k NN distances for $k = 2$ and $k = 4$ for all points

Main References

- Tan, et al., “Introduction to Data Mining”, 2nd Edition, Pearson, 2018
- M. J. Zaki and W. Meira Jr. “Data Mining and Analysis: Fundamental Concepts and Algorithms, Cambridge, 2nd Edition, 2020
- Han, J. and Kamber, M. “Data Mining: Concepts and Techniques”, 2nd edition, Morgan Kaufmann, 2006